전산화학 (2010년 1학기)

1차 시험 해답

(총 7 문제, 70 점)

1. **(총 15점, 각 5점)**

(a)

$$\begin{aligned} z = \Big(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\Big)^{40} &= \Big(\cos\Big(\frac{7\pi}{6}\Big) + i\sin\Big(\frac{7\pi}{6}\Big)\Big)^{40} \\ &= \Big(\cos\Big(40\times\frac{7\pi}{6}\Big) + i\sin\Big(40\times\frac{7\pi}{6}\Big)\Big) \\ &= \Big(\cos\Big(46\pi + \frac{2\pi}{3}\Big) + i\sin\Big(46\pi + \frac{2\pi}{3}\Big)\Big) \\ &= \cos\Big(\frac{2\pi}{3}\Big) + i\sin\Big(\frac{2\pi}{3}\Big) = -\frac{1}{2} + i\frac{\sqrt{3}}{2} \end{aligned}$$

$\therefore \mathrm{Re}(z) = -\frac{1}{2}, \qquad \mathrm{Im}(z) = \frac{\sqrt{3}}{2}.$

(b)

$$\begin{aligned} z = (-1+i)^{50} &= \sqrt{2}\Big(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\Big) \\ &= \Big(\sqrt{2}\Big(\cos\Big(\frac{3\pi}{4}\Big) + i\sin\Big(\frac{3\pi}{4}\Big)\Big)\Big)^{50} \\ &= 2^{25}\Big(\cos\Big(50\times\frac{3\pi}{4}\Big) + i\sin\Big(50\times\frac{3\pi}{4}\Big)\Big) \\ &= 2^{25}\Big(\cos\Big(36\pi + \pi + \frac{2\pi}{4}\Big) + i\sin\Big(36\pi + \pi + \frac{2\pi}{4}\Big)\Big) \\ &= 2^{25}\Big(\cos\Big(\pi + \frac{\pi}{2}\Big) + i\sin\Big(\pi + \frac{\pi}{2}\Big)\Big) \\ &= 2^{25}\Big(\cos\Big(\frac{3\pi}{2}\Big) + i\sin\Big(\frac{3\pi}{2}\Big)\Big) = -2^{25}i \end{aligned}$$

$\therefore \mathrm{Re}(z) = 0, \qquad \mathrm{Im}(z) = -2^{25}.$

(c) $z = i^i = \Big(\cos\Big(\frac{\pi}{2}\Big) + i\sin\Big(\frac{\pi}{2}\Big)\Big)^i = \big(e^{i\pi/2}\big)^i = e^{-\pi/2}$

$\therefore \mathrm{Re}(z) = e^{-\pi/2}, \qquad \mathrm{Im}(z) = 0.$

2. **(총 10점, 각 5점)** De Moivre 공식에 의해

$$(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta \tag{1}$$

한편,

$$\begin{aligned} (\cos\theta + i\sin\theta)^3 &= (\cos\theta + i\sin\theta)^2(\cos\theta + i\sin\theta) \\ &= (\cos^2\theta - \sin^2\theta + i2\sin\theta\cos\theta)(\cos\theta + i\sin\theta) \\ &= (\cos^3\theta - \sin^2\theta\cos\theta + i2\sin\theta\cos^2\theta + i\sin\theta\cos^2\theta - i\sin^3\theta - 2\sin^2\theta\cos\theta) \\ &= (\cos^3\theta - 3\sin^2\theta\cos\theta + i3\sin\theta\cos^2\theta - i\sin^3\theta) \\ &= (\cos^3\theta - 3(1-\cos^2\theta)\cos\theta + i3\sin\theta(1-\sin^2\theta) - i\sin^3\theta) \\ &= 4\cos^3\theta - 3\cos\theta + i(3\sin\theta - 4\sin^3\theta) \end{aligned} \tag{2}$$

(1) 식과 (2) 식을 비교하면,

(a) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

(b) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

임을 알 수 있다.

3. **(총 10점, 각 5점)**

(a)

$$\begin{aligned}\sinh ix = \frac{e^{ix}-e^{-ix}}{2} &= \frac{\cos ix + i\sin ix - (\cos(-ix) + i\sin(-ix))}{2} \\ &= \frac{\cos ix + i\sin ix - (\cos ix - i\sin ix)}{2} \\ &= \frac{\cos ix + i\sin ix - \cos ix + i\sin ix}{2} \\ &= i\sin ix\end{aligned}$$

(b)

$$\begin{aligned}\cosh ix = \frac{e^{ix}+e^{-ix}}{2} &= \frac{\cos ix + i\sin ix + (\cos(-ix) + i\sin(-ix))}{2} \\ &= \frac{\cos ix + i\sin ix + (\cos ix - i\sin ix)}{2} \\ &= \frac{\cos ix + i\sin ix + \cos ix - i\sin ix}{2} \\ &= \cos ix\end{aligned}$$

4. **(총 10점, 각 5점)**

(a) $m = 0$ 0이므로

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}}$$

그러므로

$$\int_0^{2\pi} d\phi \Phi_m(\phi) = \int_0^{2\pi} d\phi \frac{1}{\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi}} 2\pi = \sqrt{2\pi}.$$

(b) $m \neq 0$ 0이므로

$$\begin{aligned}\int_0^{2\pi} d\phi \Phi_m(\phi) &= \int_0^{2\pi} d\phi e^{im\phi} \\ &= \int_0^{2\pi} d\phi(\cos m\phi + i\sin m\phi) \\ &= \Big[\frac{1}{m}\sin m\phi - i\frac{1}{m}\cos m\phi\Big]_0^{2\pi} \\ &= \frac{1}{m}\Big(\sin(m2\pi) - \sin(0) - i(\cos(m2\pi) - \cos(0))\Big) \\ &= 0. \qquad (m \text{ 0이 정수0이므로}) \end{aligned} \tag{3}$$

5. **(5 점)** 직교 좌표계에서의 한 점 (x, y) 는 극 좌표계에서 다음과 같0이 변형된다.

$$x = r\cos\theta, \tag{4}$$

$$y = r\sin\theta.. \tag{5}$$

그러므로,

$$r^2 = x^2 + y^2, \tag{6}$$

$$\tan\theta = \frac{x}{y}. \tag{7}$$

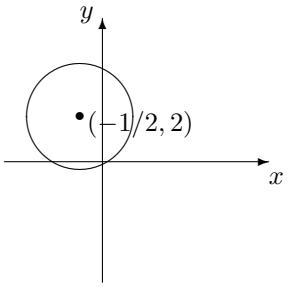
$r = 4\sin\theta - \cos\theta$ 를 다시 쓰면

$$r = 4\sin\theta \Rightarrow r^2 = 4r\sin\theta - r\cos\theta.$$

(4), (5), (6) 식을 이용하면, 다음과 같은 식을 얻을 수 있다.

$$\begin{aligned} x^2 + y^2 = 4y - x &\Rightarrow x^2 + x + y^2 - 4y = 0, \\ &\Rightarrow x^2 + x + \frac{1}{4} - \frac{1}{4} + y^2 - 4y + 4 - 4 = 0, \\ &\Rightarrow \left(x + \frac{1}{2}\right)^2 + (y-2)^2 = \left(\frac{\sqrt{17}}{2}\right)^2. \end{aligned} \tag{8}$$

(8) 식은 중심이 $(-1/2, 2)$ 이고, 반지름 $r = \sqrt{17}/2$ 인 원을 나타낸다.



6. **(총 10점, 각 5점)**

(a)

$$e^{h\nu/k_BT} \approx 1 + \frac{h\nu}{k_BT} + \frac{1}{2!}\left(\frac{h\nu}{k_BT}\right)^2 + \frac{1}{3!}\left(\frac{h\nu}{k_BT}\right)^3 = 1 + \frac{h\nu}{k_BT} + \frac{h^2}{2k_B^2T^2}\nu^2 + \frac{h^3}{6k_B^3T^3}\nu^3.$$

(b) $k_BT \gg \nu$ 이므로, $\nu/k_BT << 1$ 이고 (a)를 더욱 더 근사하면

$$e^{h\nu/k_BT} \approx 1 + \frac{h\nu}{k_BT} + \frac{h^2}{2k_B^2T^2}\nu^2 + \frac{h^3}{6k_B^3T^3}\nu^3 \approx 1 + \frac{h\nu}{k_BT} \tag{9}$$

(9) 식을 주어진 문제에 대입하면

$$\rho(\nu, T)d\nu = \frac{8\pi h}{c^3}\frac{\nu^3 d\nu}{e^{h\nu/k_BT} - 1} \approx \frac{8\pi h}{c^3}\frac{\nu^3 d\nu}{h\nu/k_BT} = \frac{8\pi h}{c^3}\frac{\nu^3 k_BT d\nu}{h\nu} = \frac{8\pi k_BT}{c^3}\nu^2 d\nu$$

7. **(총 10점, 각 5점)**

(a) $\cos\theta \approx 1 - \frac{1}{2}\theta^2$

(b) $\theta \ll 1$ 이므로 (a)를 이용하면 문제에 주어진 식은

$$-\frac{1}{\ln(\cos\theta)} \approx -\frac{1}{\ln(1 - \theta^2/2)}$$

수업 시간에 배운 $x \ll 1$ 일 때 $\ln(1-x) \approx -x$ 를 이용하면 위 식은

$$-\frac{1}{\ln(1 - \theta^2/2)} \approx -\frac{1}{-\theta^2/2} = \frac{2}{\theta^2}$$